Cycle Test 1: worked solutions

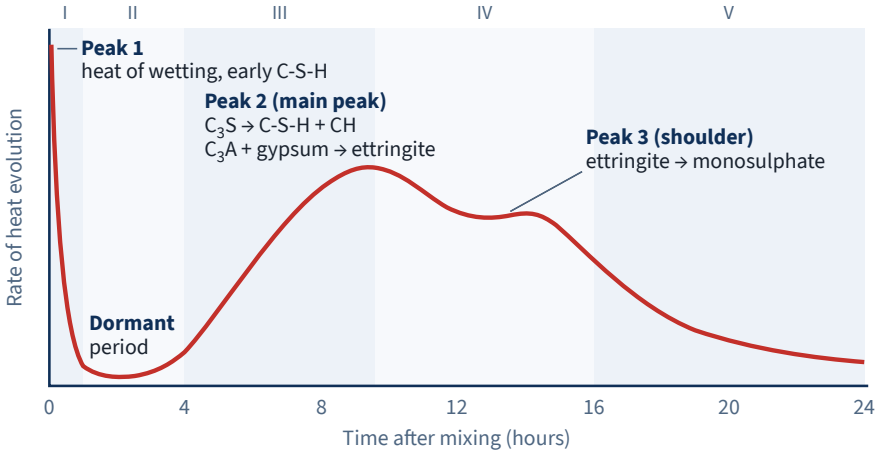
20 Aug 2025 · 20 marks · 60 min · CEPC11

Part A (1 mark each)

Q1

1 mark

> Draw the hydration evolution pattern indicating peak reaction over the first 24 hours.



- **Peak 1** (first minutes): heat of wetting, dissolution of ions.
- **Dormant period** (about 1–3 h): very slow reaction; concrete stays workable.
- **Peak 2, the main (peak) reaction** at about **8–10 h**: C_3S hydrates rapidly to C-S-H + CH; setting occurs.
- **Peak 3** (shoulder, 12–20 h): ettringite (AF_t) converts to monosulphate (AF_m) once gypsum is used up.

Q2

1 mark

Match the cement type with its property.

	Cement type		Property
A	Type II	I	Lower C_3A
B	Type III	II	Slower development of strength
C	Type IV	III	Marginally lower C_3S and C_3A
D	Type V	IV	Marginally higher C_3S

Cement type	Matches	Why
A. Type II (modified)	**III. Marginally lower C_3S and C_3A**	Moderate heat, moderate sulfate resistance
B. Type III (rapid hardening)	**IV. Marginally higher C_3S**	Plus higher fineness → high early strength
C. Type IV (low heat)	**II. Slower development of strength**	Low C_3S, high C_2S
D. Type V (sulfate resisting)	**I. Lower C_3A**	$C_3A \leq 4\%$ resists sulfate attack

Answer: A–III, B–IV, C–II, D–I.

Q3

1 mark

How does the shrinkage reducing admixture control the shrinkage cracks?

When concrete dries, water is lost from the **capillary pores (about 2.5–50 nm)**. **Menisci** form, and the **surface tension of the water** pulls inward on the pore walls; this tension causes drying shrinkage and, where restrained, cracking.

A **shrinkage reducing admixture (SRA)**, e.g. **polyoxyalkylene alkyl ether**, **lowers the surface tension of the pore water**. Lower surface tension means a smaller capillary pull, so **less shrinkage strain and lower tensile stress**, and fewer cracks. Dosage is high: about **2–4% by weight of cement**.

Q4

1 mark

> Brief any two issues associated with the quality of water in concrete.

Any two of:

1. **Chlorides and sulfates** (brackish water): chloride above about 500 ppm causes **corrosion of reinforcement**; SO_3 above about 1000 ppm causes **sulfate attack**.
2. **Organic matter / humic acids, algae**: interfere with setting and hardening; algae entrain air and **reduce strength**.
3. **High sodium or potassium content**: can cause **alkali–aggregate reaction**, even in drinkable water.
4. **Iron or organic matter in curing water**: **staining** of the surface.
5. **pH outside 6–8**, or **excess silt** (over 2000 ppm), affects setting and strength.

Part B (4 marks each)

Q5

4 marks

> State the importance of the kiln reaction in the production of cement.

The kiln converts an inert mixture of limestone and clay into **clinker**, made of **reactive compounds that can hydrate and gain strength**. Without these reactions there is no cement.

Temperature	Reaction	Why it matters
Up to 700 °C	Water removed; clay minerals activated	Prepares silica and alumina to react
700–900 °C	**Decarbonation**: $CaCO_3 \rightarrow CaO + CO_2$	Releases the lime (CaO) needed for all compounds
900–1200 °C	**Belite (C_2S)** forms	Gives later-age strength
Above 1250–1300 °C	**Liquid phase** forms; $C_2S + CaO \rightarrow$ **alite (C_3S)**	C_3S gives early strength; the melt speeds the reaction

Temperature	Reaction	Why it matters
Cooling	Melt becomes glass with C_3A and C_4AF	Fast cooling prevents C_3A crystallizing (setting problems) and C_3S breaking down

Why the kiln reaction is important:

- It decides the **compound composition** (C_3S, C_2S, C_3A, C_4AF) and therefore **strength gain, heat of hydration and durability**.
- **Incomplete burning leaves free lime**, which makes the cement **unsound** (delayed expansion). This is controlled by keeping the **LSF at 92–98%**.
- The **burning temperature and cooling rate** decide crystal size and reactivity (modern quick burning gives 10–40 µm alite).
- Alkali sulfates deposited on clinker surfaces affect early setting.
- The kiln is the most **energy- and CO_2-intensive** step, so it controls the cost and environmental impact of cement.

Q6

4 marks

> a) Brief the significance of physical properties of aggregate on the fresh properties of concrete.
>
> b) Suggest the application for different gradings of aggregate in concrete construction.

a) Physical properties and fresh concrete (aggregate is about 75% of the volume, so it controls fresh behaviour):

- **Shape**: rounded particles give better **workability**; angular, flaky or elongated particles increase **water demand** and harshness (angularity number 0 → 11 means less workable). Flaky particles above 10–15% are undesirable.
- **Surface texture**: rough surfaces need more paste and water than smooth ones.
- **Size and grading**: well-graded aggregate has fewer voids, so less paste is needed; larger maximum size reduces surface area and water demand. Poor grading causes **segregation and bleeding**.

- **Fineness of sand (FM)**: very fine sand raises water demand; very coarse sand makes the mix harsh and prone to bleeding.
- **Absorption and moisture state**: dry aggregate absorbs mixing water (stiffer mix, **slump loss**); wet aggregate adds water (higher w/c). Batching must correct to SSD.
- **Bulking of sand** affects volume batching.
- **Fines, silt and clay**: raise water demand.

b) Gradings and their applications:

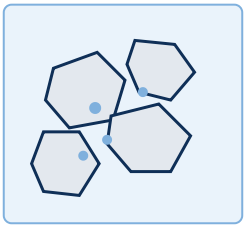
Grading	Characteristics	Application
Dense / well graded	Wide range of sizes, low voids, high density	General **structural concrete**, pavements; economical (least paste) and durable
Gap graded	Middle sizes missing; easy to compact	**Exposed aggregate** finishes, **preplaced aggregate** concrete, low-workability mixes compacted by vibration
Open graded	Few fines, high voids, permeable	**Pervious (no-fines) concrete**, drainage layers, porous pavements
Uniform graded	Narrow size range	Lightweight / no-fines concrete, filter media

Q7

4 marks

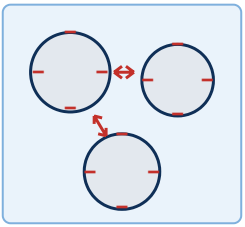
> Compare and differentiate the mechanism of superplasticizers and viscosity modifying admixture.

Without admixture



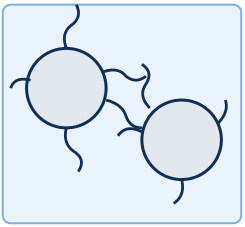
grains flocculate and trap water (blue) inside flocs

SNF / SMF / lignosulphonate



adsorbed molecules give the grains like (negative) charges: **electrostatic repulsion**

PCE (polycarboxylate)



long side chains stick out and physically keep grains apart: **steric hindrance**

Either way the grains are dispersed, the trapped water is released, and the mix flows more for the same water content.

PCE also keeps working longer (better slump retention) and allows up to ~40% water reduction.

	Superplasticizer (HRWR)	Viscosity modifying admixture (VMA)
Purpose	**Increase flow** / reduce water (15–40%)	**Increase viscosity and cohesion** of very fluid concrete; anti-washout under water
Chemicals	SNF, SMF, lignosulphonates (high dose), **PCE**	Long-chain water-soluble polysaccharides: **cellulose ethers, welan, xanthan, diutan gum, starch**
Acts on	**Cement particles**	**Mixing water**
Mechanism	Adsorbs on cement grains → **electrostatic repulsion** (lower zeta potential; SNF/SMF) and **steric hindrance** (PCE side chains). Releases water trapped in flocs	**Adsorption**: binds part of the free water and swells. **Association**: attractive forces between adjacent chains form a gel. **Intertwining**: chains entangle at low shear (shear thinning at high shear)
Effect on viscosity	Lowers yield stress and viscosity	Raises viscosity; reduces bleeding and segregation
Dosage	0.7–1.0% by weight of cement	0.03–0.08% by weight of cement

Together they are used in **self-compacting concrete**: the SP gives flow, the VMA keeps the mix stable.

Q8

4 marks

> a) Explain pozzolanic reaction and its benefits.
>
> b) Explain any two merits and challenges associated with any two types of SCMs with justification.

a) Pozzolanic reaction

A **pozzolan** is a siliceous or aluminous material with little or no cementing value by itself which, **in finely divided form and in the presence of moisture, reacts with calcium hydroxide** at ordinary temperature to form cementitious compounds (ASTM C595).

$$Ca(OH)_2 + \text{reactive } SiO_2 \text{ (or } Al_2O_3) + H_2O \rightarrow \text{C-S-H (or C-A-H)}$$

Benefits:

- **Consumes CH**, the weakest and most easily attacked hydration product.
- Produces **additional C-S-H**, so long-term strength increases.
- **Pore refinement** and **interface (ITZ) refinement** → lower permeability.
- Better **durability**: resistance to sulfates, chlorides and ASR.
- **Lower heat** of hydration; less cement → lower cost and CO_2.
- The reaction is **slow** (gains strength later) and is accelerated by alkalis and gypsum.

b) Two SCMs: merits and challenges

SCM	Merits	Challenges
Fly ash	Spherical particles → **ball-bearing effect, better workability**, less bleeding. **Lower heat**, higher long-term strength, reduced ASR (dilution of alkalis). Cheap industrial by-product	**Slow early strength** (formwork removal delayed). **Unburnt carbon (LOI)** interferes with air entrainment. **Variable quality**; uneconomical beyond 40–50 km from the power plant. Higher creep and shrinkage if paste volume rises

SCM	Merits	Challenges
Silica fume	Extremely fine (BET 13,000–30,000 m^2/kg): strong **pozzolanic + filler** effect → very high strength, **very low permeability**, lower chloride ingress and corrosion, better abrasion resistance	**High water demand**, so a superplasticizer is always needed; **sticky mix**, slump loss; almost **no bleeding → plastic shrinkage** cracks unless cured promptly; **costs ~10× cement**; poor fire performance
GGBS	Better durability, especially in **marine** works; improved ultimate strength	Needs alkali/sulfate activation; **slow early strength**